

Figure Guide

What every diagnostic figure shows, how to read it, and what it decides

Volatility and tail-risk forecasting — six Tier-1 equity indices

What these figures are — and are not

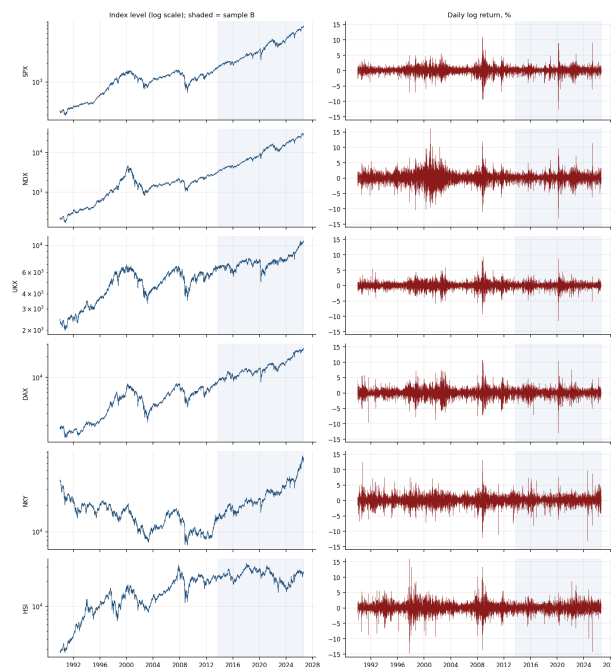
These ten figures are **diagnostics, not data**. They were generated from the dataset during exploratory analysis to test specific questions, and they exist as evidence for the cleaning and modelling decisions that follow. Nothing in the pipeline reads them; deleting them would not affect the dataset. `10_SCRIPTS/19_eda_figures.py` regenerates all ten in about thirty seconds.

Two of them are load-bearing rather than illustrative. Figure 06 is the reason the Nikkei's 2016-17 realized measures were nulled, and figure 09 is the reason a symmetric GARCH is not an acceptable baseline. If either were wrong, the corresponding decision would have to be revisited.

Likely use in the paper

Figure	Where	Why
02, 03, 09	Main text	The stylised facts that motivate the model class
07	Methodology	Justifies the EVT threshold choice
04, 08	Main text or appendix	Variance risk premium; cross-index dependence
01, 05, 06, 10	Appendix / supplementary	Data provenance and quality evidence

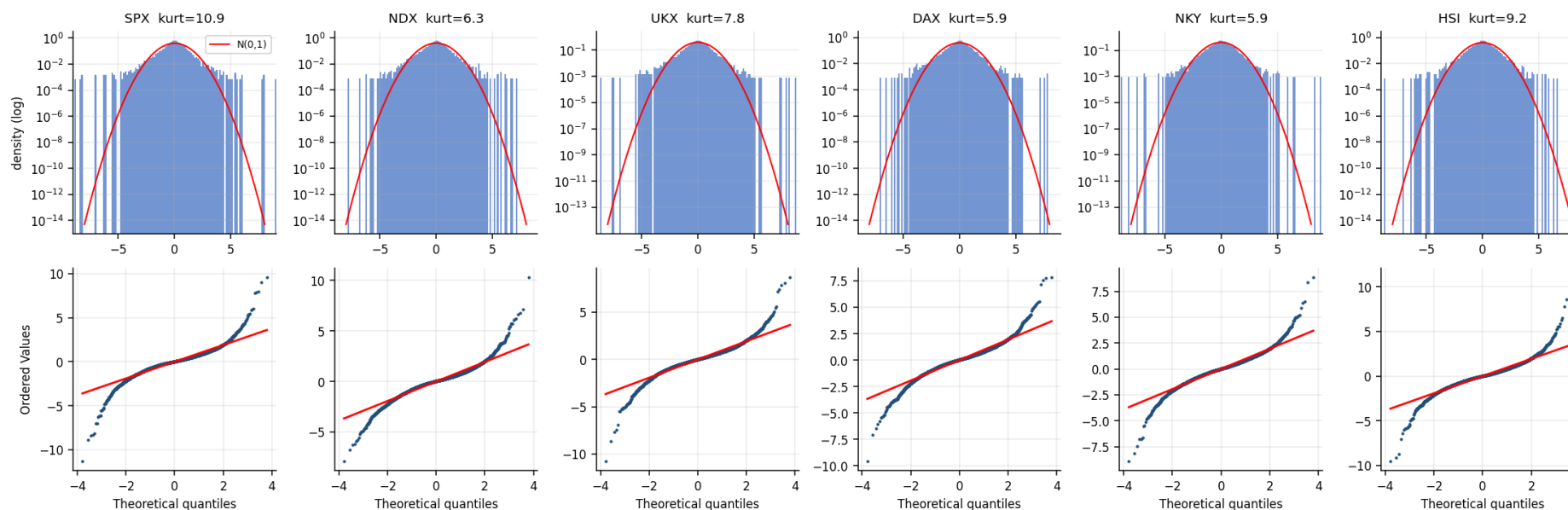
Figure 01 — Price and return timeline



What it plots	Twelve panels. Left column: the closing level of each index on a LOG scale, so equal vertical distances are equal percentage moves. Right column: the daily log return in per cent, capped at +/-16 for readability. The blue shaded band on both is sample B.
How to read it	Read the left column for trend and the right for risk. On a log scale a straight line is constant compound growth. In the return panels look for VERTICAL THICKENING - bands of large moves clustered together - rather than for individual spikes.
What our data shows	All six show the same three thickenings: 2000-02, 2008-09 and 2020. The shaded band makes the central compromise visible at a glance - sample B starts after the GFC, so the realized-measure models never see 2008, while the daily-only models do.
What it decides	Confirms each series is the index it claims to be, with no level breaks, splits or stitching errors, and locates the stress episodes that the regime analysis will use.
Caveat	The +/-16% clip hides the exact height of the very largest moves. Read those off the Worst_Days table, not this figure.

Figure 02 — Return distribution and normal QQ plots

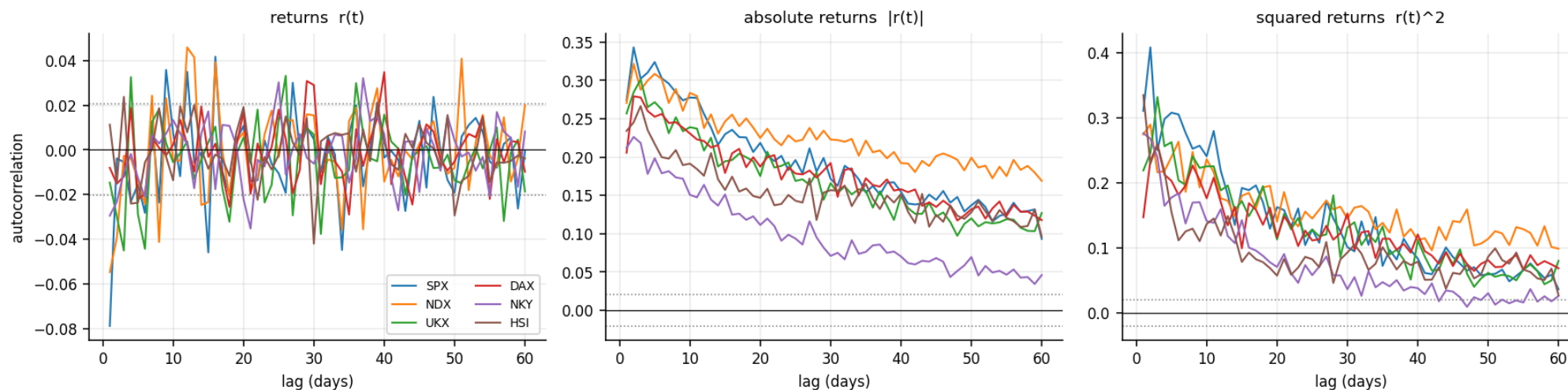
Standardised daily returns vs the normal distribution: density on a log scale (top) and normal QQ plot (bottom). Tail departure is the point.



What it plots	Top row: histogram of standardised returns with the standard normal density in red, plotted with a LOGARITHMIC y-axis. Bottom row: normal quantile-quantile plots - empirical quantiles against theoretical normal quantiles, with the red line showing where points would fall if returns were Gaussian.
How to read it	The log y-axis is the whole point: on a linear scale the tails are invisible because the density is tiny there. On a log scale a Gaussian is a downward parabola, so anything that sits ABOVE the red curve out in the wings is excess tail mass. In the QQ plot, points bending BELOW the line on the left and ABOVE on the right mean both tails are fatter than normal; the further they bend, the fatter.
What our data shows	Every index bends away from the line at both ends, and the left end bends further than the right - fat tails plus negative skew. Excess kurtosis runs 5.9 to 10.9. The most extreme points sit five to ten standard deviations out, which under a Gaussian would be essentially impossible.
What it decides	Rules out a Gaussian innovation and motivates the EVT stage. Together with the Hill estimates it is the visual half of the argument that a normal-innovation GARCH will understate 1% VaR.
Caveat	Standardising by the FULL-SAMPLE mean and standard deviation, as here, mixes calm and crisis periods, so some of the apparent fat-tailedness is volatility clustering rather than true unconditional tail weight. That is exactly why the EVT stage is applied to GARCH residuals and not to these raw returns.

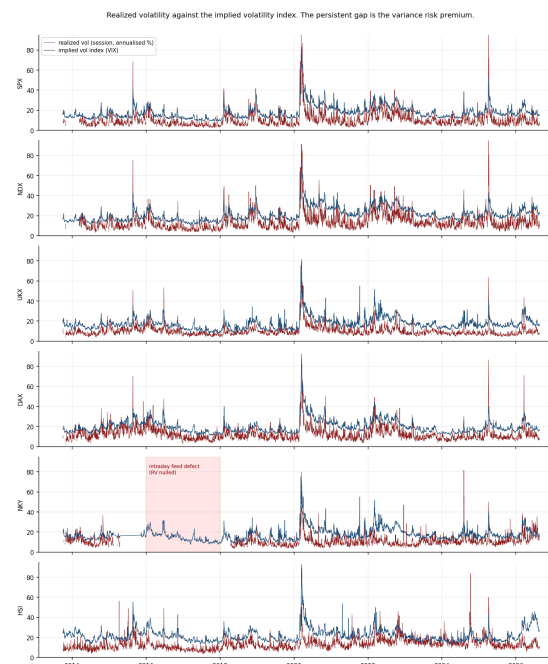
Figure 03 — Autocorrelation of returns, absolute returns and squared returns

Volatility clustering: returns are near-white, but $|r|$ and r^2 stay autocorrelated for months. This is the precondition for a GARCH model.



What it plots	Three panels sharing a lag axis of 1 to 60 trading days, with all six indices overlaid. Left: autocorrelation of the return itself. Middle: of the absolute return. Right: of the squared return. The dotted grey lines are the approximate 95% confidence band around zero.
How to read it	Compare the three panels against each other. If a series were independent, all three would sit inside the band. What you should see is the left panel near zero and the other two clearly and persistently positive.
What our data shows	Exactly that. Returns are close to white noise - a small negative first-order autocorrelation and little else. Absolute and squared returns stay well above the confidence band out to sixty days and decay slowly rather than geometrically.
What it decides	This is the precondition for GARCH. Returns are nearly unpredictable in the MEAN but strongly predictable in the VARIANCE, which is the entire premise of conditional volatility modelling. The slow, non-geometric decay is also the first hint of long memory, which the GPH estimates confirm.
Caveat	The confidence band assumes independence and is therefore too narrow for the $ r $ and r^2 panels. Treat those bands as indicative; the formal evidence is the Ljung-Box and ARCH-LM tests, which reject at $p < 1e-16$.

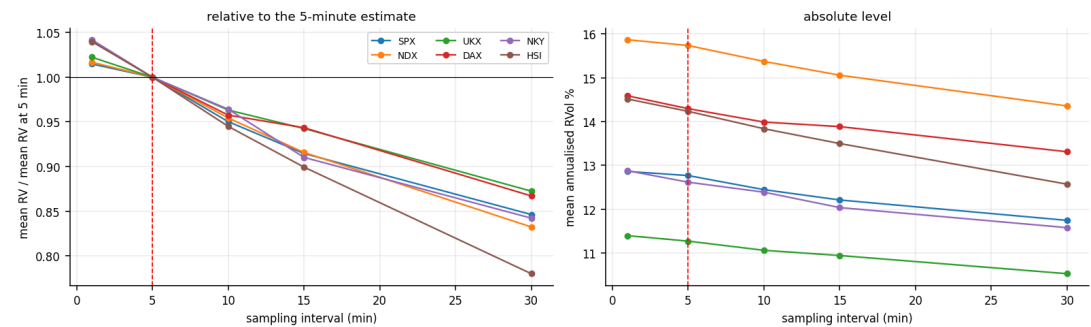
Figure 04 — Realized volatility against the implied volatility index



What it plots	One panel per index. Dark red is realized volatility from the 5-minute intraday data, annualised into per cent. Dark blue is that market's implied-volatility index. Both on the same axis, capped at 95.
How to read it	Look at the GAP between the two lines, not just their co-movement. Implied volatility sits persistently above realized volatility; that wedge is the variance risk premium, the compensation option sellers earn. Then look at the SPIKES: implied typically jumps first and overshoots.
What our data shows	The premium is positive nearly everywhere and widens sharply in stress. The red block on the Nikkei panel marks 2016-17, where the intraday feed is broken and realized volatility has been nulled - the red line simply stops.
What it decides	Justifies including VRP and the implied-volatility level as quantile-regression predictors, and gives the visual context for the screening result that implied volatility beats every realized measure for the 1% tail.
Caveat	The two series are NOT on identical footing. Realized volatility here covers the cash session only, while the implied index prices a full 30 calendar days including overnight and weekend risk. Part of the visible gap is that mismatch rather than a risk premium - use ScaleFactor_HL before quantifying the premium.

Figure 05 — Volatility signature plot

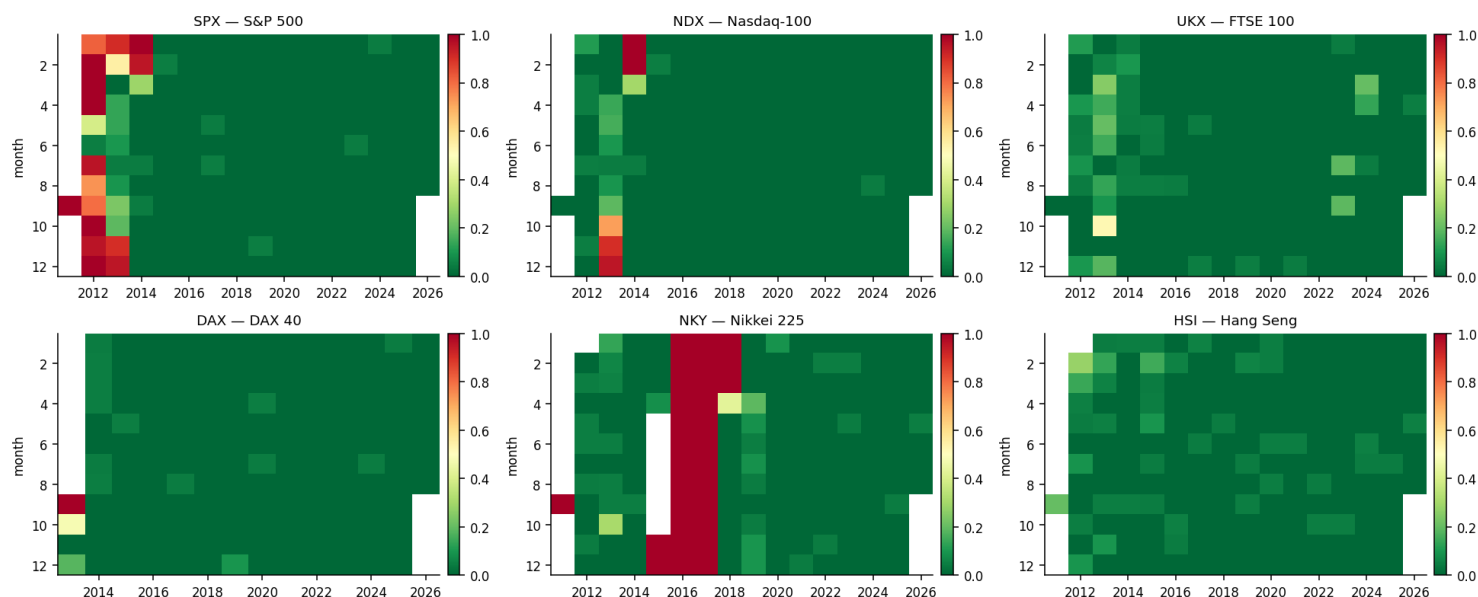
Volatility signature plot. RV declines mildly and monotonically with the sampling interval — about 15-20% in variance between 1 and 30 minutes — rather than showing the sharp high-frequency blow-up of noisy trade data. The 5-minute grid (red) sits within a few percent of the 1-minute estimate.



What it plots	Left: mean realized variance at 1, 5, 10, 15 and 30-minute sampling, divided by the 5-minute value so all six indices share one axis. Right: the same in absolute annualised volatility per cent. The red dashed line marks the 5-minute grid actually used.
How to read it	This is the standard microstructure-noise diagnostic. Sampling too finely makes RV pick up bid-ask bounce and quote noise instead of true volatility, which INFLATES it. The textbook picture is a sharp rise as the interval shortens toward zero, flattening out somewhere around 5 to 30 minutes. You choose the coarsest frequency that is still on the flat part.
What our data shows	The profile declines mildly and monotonically - roughly 15 to 20% in variance between 1 and 30 minutes - with no sharp high-frequency blow-up. The 1-minute to 5-minute ratio is only 1.06 to 1.09, against 1.3 to 2.0 typical of trade data. The feed is a smoothed CFD quote, so it carries little microstructure noise.
What it decides	Justifies the 5-minute sampling choice: it sits within a few per cent of the 1-minute estimate, so almost nothing is lost, while remaining the literature standard and immune to the noise objection.
Caveat	Low noise is not automatically good news. A smoothed quote may also DAMP genuine high-frequency variation, which would bias RV downward relative to a trade-based estimator. This is disclosed as a limitation, not presented as a strength.

Figure 06 — Intraday session quality by month

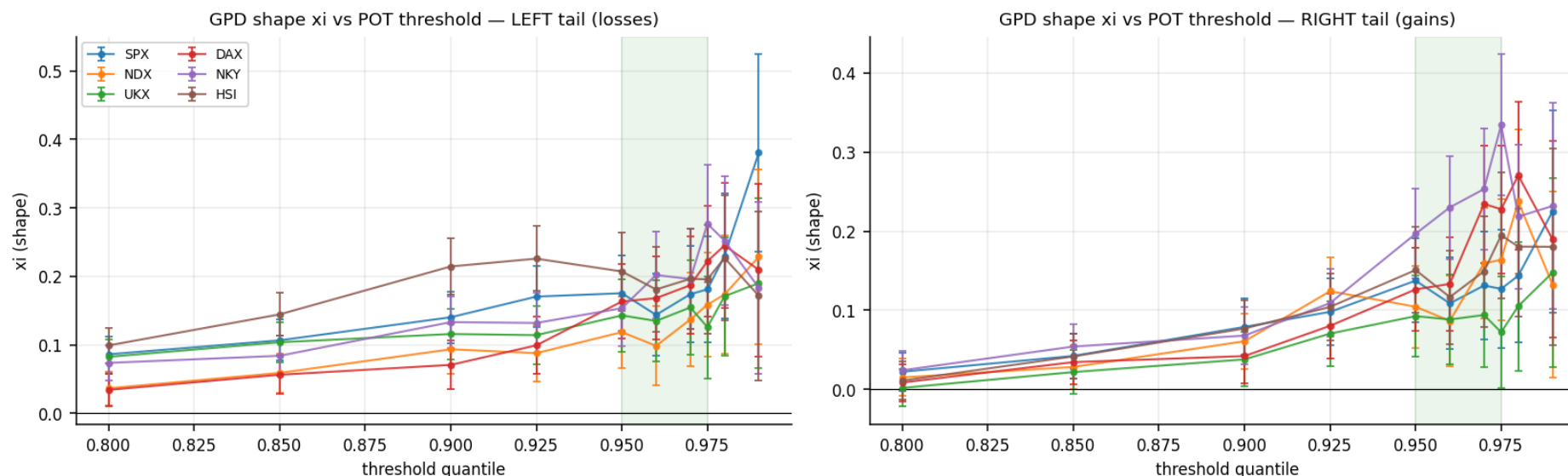
Share of each month's intraday sessions classified DEFECT (feed failure). Green = clean, red = wholly unusable. The NKY 2016-17 block is the defect that drove the cleaning rule; white = no data cached for that month.



What it plots	One panel per index, months down the vertical axis and years across the horizontal. Colour is the SHARE of that month's sessions classified DEFECT: green is clean, red is wholly unusable. White means no data was cached for that month.
How to read it	Scan for red blocks. Isolated red cells are ordinary one-off feed interruptions. A solid red rectangle spanning many months is a systematic outage and is a completely different problem.
What our data shows	The Nikkei has a solid red block covering the whole of 2016 and 2017 - the Dukascopy feed carries no bars at all in the 09:00 local hour for those two years. The S&P; shows a smaller cluster in 2012-13. Everything else is green.
What it decides	This is the figure behind the single most consequential cleaning decision in the project: nulling the Nikkei's realized measures for 2016-17. Because intraday volatility is U-shaped, losing the opening hour biases RV downward by far more than the missing time fraction, measured at 24% low against the Parkinson benchmark.
Caveat	An earlier version of this figure averaged a three-level class code and passed it through a three-colour palette, which silently rendered a month that was half broken as if it were a half-day. It now plots the defect SHARE on a continuous scale. If you regenerate it, do not revert to the categorical version.

Figure 07 — Generalised Pareto threshold stability

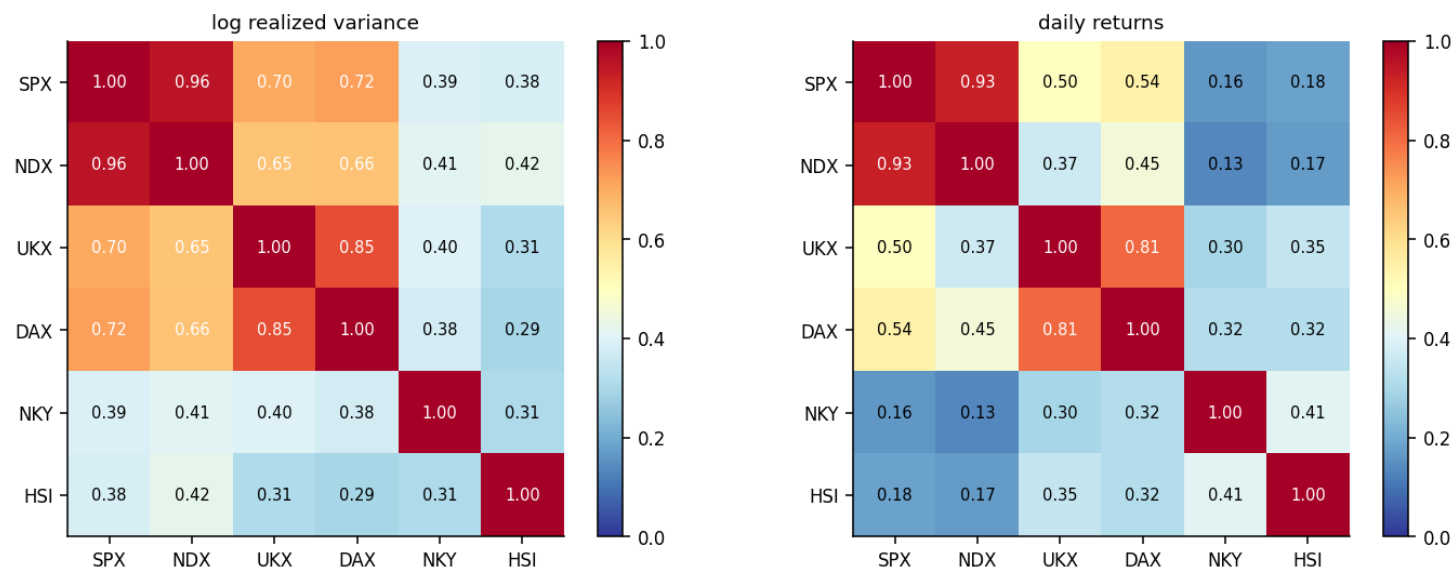
Peaks-over-threshold stability. $\xi > 0$ means a heavy (Frechet) tail even AFTER volatility standardisation. Shaded = the recommended 95-97.5% region.



What it plots	Two panels, left tail and right tail. The horizontal axis is the threshold expressed as a quantile from 0.80 to 0.99; the vertical axis is the fitted GPD shape parameter ξ with asymptotic error bars. One line per index. The green band marks the recommended 95 to 97.5% region.
How to read it	You are looking for a PLATEAU. Below the right threshold the GPD limit theory has not taken hold and ξ drifts as the threshold rises; above it, so few exceedances remain that the estimate becomes noisy and the error bars widen. The correct threshold is the lowest one inside the stable stretch, because that keeps the most data.
What our data shows	ξ is positive at every threshold for every index - a heavy Frechet-domain tail. It rises from around 0.03-0.10 at the 80th percentile and settles around 0.15-0.25 from roughly the 95th, before becoming erratic above the 98th where fewer than 190 exceedances are left.
What it decides	Sets the peaks-over-threshold cutoff at the 95th to 97.5th percentile, giving 230 to 460 exceedances per index. Critically, ξ stays positive AFTER volatility standardisation, which is precisely the McNeil-Frey argument that a GARCH filter alone does not remove the tail and a second EVT stage is needed.
Caveat	This is fitted to returns standardised by a rolling 252-day volatility, used as a stand-in because no GARCH had been fitted at the EDA stage. Genuine GARCH residuals will differ. Treat the band as a starting expectation and REDO this plot on the real residuals.

Figure 08 — Cross-index correlation

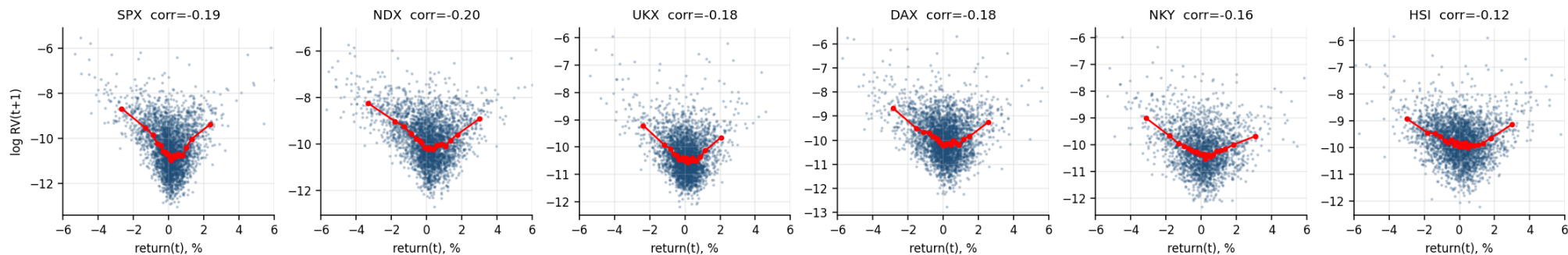
Cross-index correlation. The US and European blocks are tight; Asia is nearly uncorrelated with the US on the SAME calendar date because the sessions do not overlap.



What it plots	Two heatmaps on the balanced sample. Left: correlation of log realized variance between each pair of indices. Right: correlation of daily returns. Warmer colours are higher.
How to read it	Look for BLOCKS. Groups of indices that move together will show as bright squares along the diagonal. Then compare the two panels - volatility and returns need not share the same structure.
What our data shows	Two tight blocks: the US pair at 0.96 and the European pair at 0.85. Asia is largely detached, and Asia-to-US return correlation is only about 0.13-0.18. The first principal component of the log-RV matrix explains 62% of the variation.
What it decides	Establishes that the six indices are not six independent experiments - a pooled test that treats them as such will overstate its own significance. Supports using Hansen's Model Confidence Set, which handles dependence across the compared series.
Caveat	THE LOW ASIA-US CORRELATION IS A TIMING ARTEFACT, NOT ECONOMIC INDEPENDENCE. Tokyo and Hong Kong close before New York opens, so a US shock on day t reaches Asia on day $t+1$. Do not interpret this figure as evidence that Asian markets are decoupled.

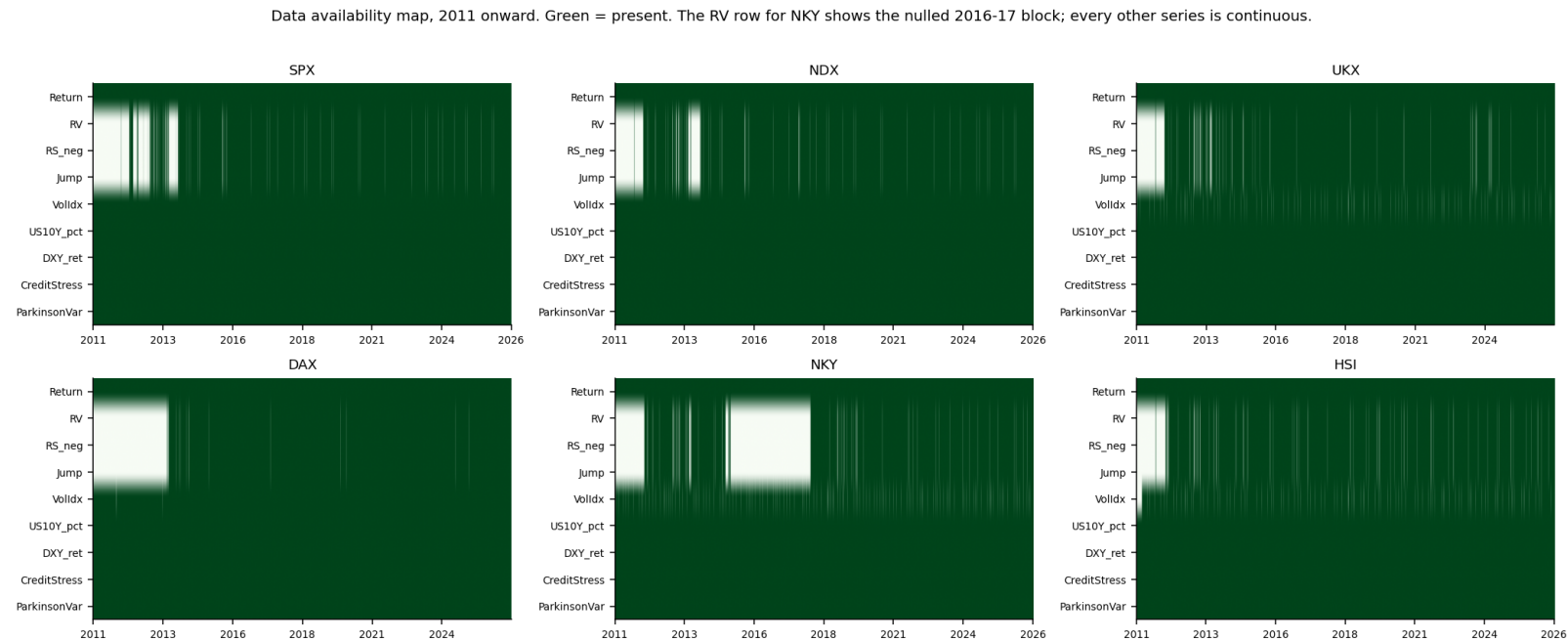
Figure 09 — News impact curve

News impact: tomorrow's realized variance against today's signed return. The red binned mean is asymmetric — negative shocks raise volatility more. This is why GJR/EGARCH is required over plain GARCH.



What it plots	One scatter per index: today's signed return in per cent on the horizontal axis against TOMORROW's log realized variance on the vertical. The red line joins the means of twenty equal-count bins, so it traces the average volatility response to a shock of each size.
How to read it	The red line is the empirical news impact curve. Under a symmetric GARCH it would be a symmetric V centred on zero: a -2% day and a +2% day would raise tomorrow's variance equally. Compare the steepness of the left arm against the right.
What our data shows	For all six the left arm is visibly steeper than the right. A negative shock raises tomorrow's volatility more than a positive shock of the same size. The correlation between today's signed return and tomorrow's log RV runs -0.12 to -0.20.
What it decides	This is the evidence that PLAIN GARCH(1,1) IS MISSPECIFIED for this data. A symmetric model cannot reproduce an asymmetric news impact curve. GJR-GARCH or EGARCH is required, and the formal Engle-Ng sign-bias test rejects on all six indices.
Caveat	The binned mean is sensitive to the tails, where few observations sit, so the extreme ends of the red line are less reliable than the middle. The x-axis is clipped at +/-6% and a handful of points lie outside it.

Figure 10 — Data availability map



What it plots	One panel per index, 2011 onward. Each row is a column of the dataset, each horizontal position a trading day. Green means the value is present, white means missing.
How to read it	Read across each row for continuity. A clean row is solid green. Gaps show as white vertical stripes, and a systematic outage as a wide white block.
What our data shows	The RV, RS_neg and Jump rows for the Nikkei show a clear white block over 2016-17 - the nulled defect period. Return, VolIdx and the range and macro rows are continuous everywhere, which is exactly the intended outcome: the intraday failure removed the realized measures without touching the daily observations.
What it decides	Documents honest missingness. It is the visual proof that rows were not deleted when the intraday feed failed - only the affected columns were nulled - so the return series that GARCH-EVT and quantile regression estimate on remains complete.
Caveat	Green means PRESENT, not VALID. A realized measure can be present and still fail the RV_Valid gate on a half-day. Always filter on RV_Valid rather than on notna().